Two worked examples, drawn step by step

every box, every centre, every drop — read out of a real run

Abstract

Two instances are followed to the end, one with two variables and one with three. Every figure is generated by `docs/figs/regenerate.py`, which re-runs the search and reads the boxes, maximisers and centres out of it: no picture here was drawn by hand, and none can drift away from the code. The smaller example is the one the published method is stated on, so the two can be compared line for line; the larger one is chosen because its four steps exhibit *every* branch of the algorithm — a repair, a maximiser that is already efficient, a box dropped on its bound, and the early exit.

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1 How to read the figures

Each step gets one picture of **criterion space** (Z_1, Z_2) — not of decision space, because that is where the algorithm keeps its unexplored region. In every picture:

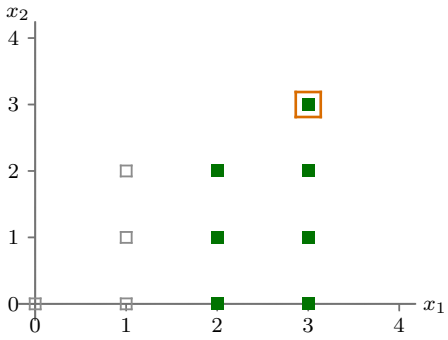
■ efficient point □ dominated point □ the box being settled ■ what the cut removes □ the children left ∘ x_B ∘ the centre cut on

The solid blue rectangle is the box currently being settled; the dashed red rectangles are the children it is about to be split into, drawn on the same axes so the split can be seen rather than described. A box open on a side extends to the frame.

2 Example A: the published instance, $n = 2$

$$\max \Phi(x) = \frac{x_1 + x_2 - 1}{5x_1 + x_2 - 1} \quad \text{over} \quad E(P_D) \text{ of } \begin{cases} Z_1(x) = x_1 - 2x_2 \\ Z_2(x) = -x_1 + 4x_2 \end{cases} \qquad D : \begin{cases} -2x_1 + x_2 \leqslant 0 \\ 6x_1 + x_2 \leqslant 21 \\ -2x_1 + 4x_2 \leqslant 6 \\ x \in \mathbb{Z}^2_+ \end{cases}$$

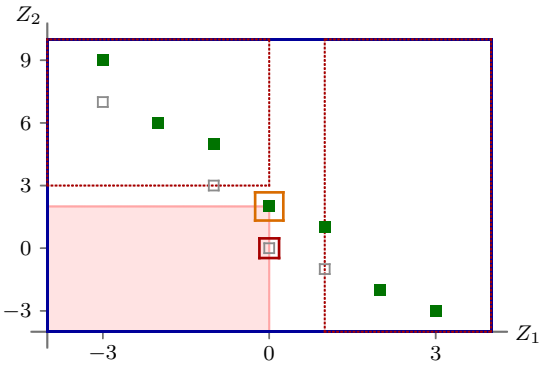
D holds 11 points, of which 7 are efficient.



Decision space. The eleven feasible points; the seven efficient ones filled, the optimum of (P_E) ringed. Note that the efficient set is not a face and not convex — there is nothing here to hand a solver, which is the difficulty (P_E) starts from.

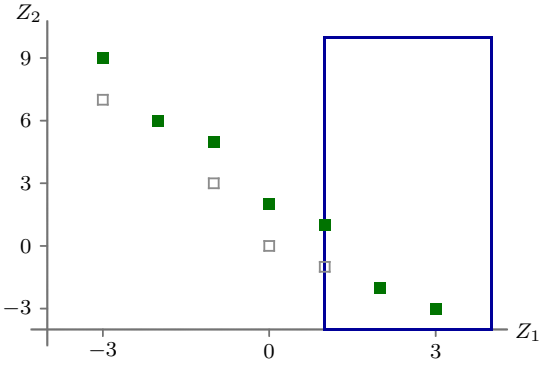
Where the answer is, and why it is not obvious. Φ is maximised over all of D at $x = (0, 0)$, with $\Phi = 1$. That point is *dominated*, so it is not a feasible answer to (P_E). The true optimum is $\Phi = 5/17 \approx 0.294$ at $(3, 3)$ — barely a third of the unconstrained value. An implementation that forgot the efficiency test would return 1 and look entirely plausible.

Step 1 — the root box



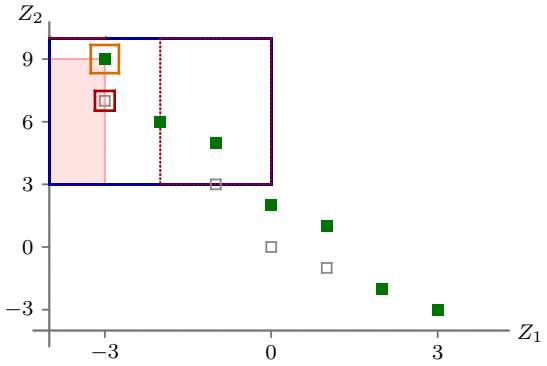
The box is all of $\mathbb{R}^2$, so the sub-problem is $\max \Phi$ over D itself: $x_B = (0, 0)$ with $\Phi = 1$. The test returns $\theta > 0$ — $(0, 0)$ is dominated — so its value is discarded rather than banked. The centre is the efficient point $(2, 1)$, at $Z = (0, 2)$. $Q(2, 1)$ gives $\Phi = 1/5$, the first incumbent. The split removes $\{Z \leqslant (0, 2)\}$ and leaves two children: $Z_1 \geqslant 1$, and $Z_2 \geqslant 3$ with $Z_1 \leqslant 0$.

Step 2 — a box dropped on its bound



$Z_1 \geqslant 1$. The sub-problem cannot beat the incumbent 1/5, so the box is discarded *whole* — it is never split, and every descendant it would have had is gone with it. This is the mechanism that a good incumbent feeds, and the reason the search is worth seeding.

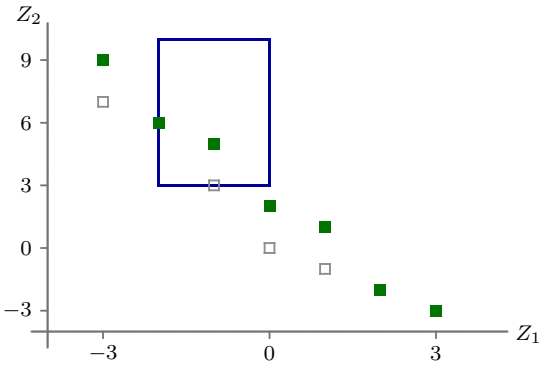
Step 3 — the optimum arrives, from a dominated maximiser

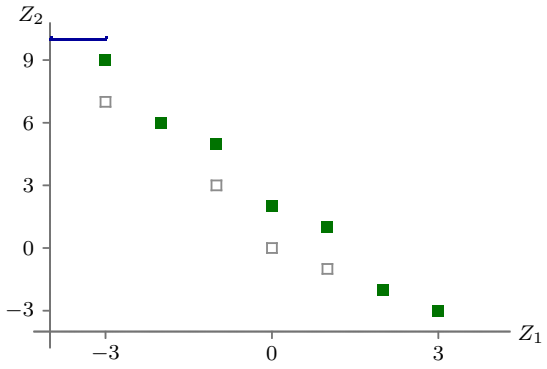


$Z_1 \leqslant 0$, $Z_2 \geqslant 3$. Here $x_B = (1, 2)$ with $\Phi = 1/3$ — larger than the incumbent, but the test rejects it: $(1, 2)$ is dominated. The centre is $(3, 3)$, and $Q(3, 3) = 5/17$, which becomes the incumbent and is, in the end, the answer.

Note what would have gone wrong twice over. Taking $\Phi(x_B) = 1/3$ as the incumbent would have been taking the value of a dominated point, and $1/3 > 5/17$: the cutoff would then have discarded the box holding the true optimum.

Steps 4 and 5 — the rest is proof





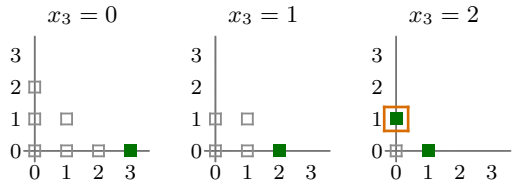
Neither child can beat 5/17; both are dropped. The list empties and the answer is proved optimal. **Five sub-problems in total**, against a feasible set of eleven points and an efficient set of seven — the method never enumerated $E(P_D)$, which is the whole point of it.

3 Example B: three variables, and every branch exercised

$$\max \Phi(x) = \frac{-2x_1 + 4x_2 - 3x_3 + 1}{x_1 + 2x_2 + x_3 + 1} \quad \text{over} \quad E(P_D) \text{ of } \begin{cases} Z_1(x) = 4x_1 - x_2 + x_3 \\ Z_2(x) = 2x_2 + 2x_3 \end{cases}$$

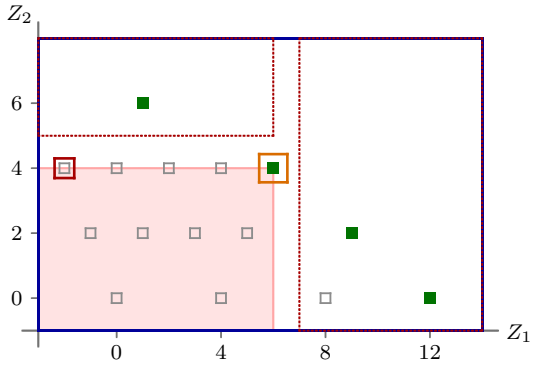
$$D: \quad x_1 + x_2 + 2x_3 \leqslant 5, \qquad 2x_1 + 3x_2 + x_3 \leqslant 6, \qquad 0 \leqslant x_j \leqslant 3, \quad x \in \mathbb{Z}^3.$$

D holds 15 points; exactly 4 are efficient, at $Z = (1, 6), (6, 4), (9, 2), (12, 0)$.



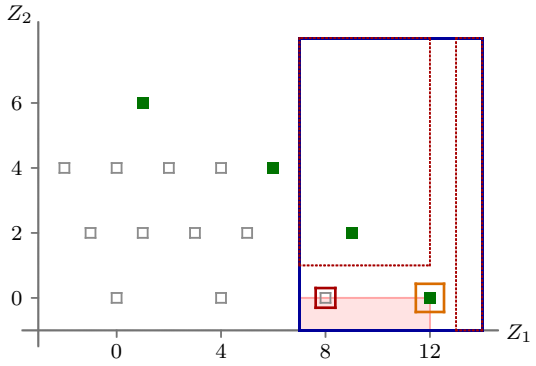
Decision space, as three slices. With $n = 3$ the lattice is drawn one x_3 at a time. The efficient points are filled and the optimum ringed; they lie in different slices, which is what makes the set awkward in decision space and tidy in criterion space — in the figures below all four sit on a single staircase.

Step 1 — repair, and the first incumbent



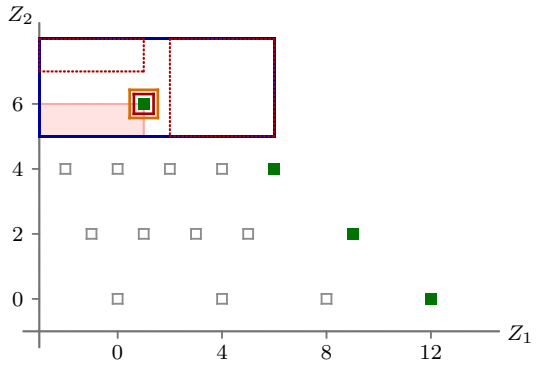
Root box. $x_B = (0, 2, 0)$ with $\Phi = 9/5$, the largest value Φ takes anywhere on D — and dominated, so again not an answer. The centre is $(1, 0, 2)$ at $Z = (6, 4)$, and Q there gives $\Phi = -7/4$. That is the incumbent: *negative*, and far below the $9/5$ just seen. The split leaves $Z_1 \geqslant 7$ and $(Z_2 \geqslant 5, Z_1 \leqslant 6)$.

Step 2 — a second repair



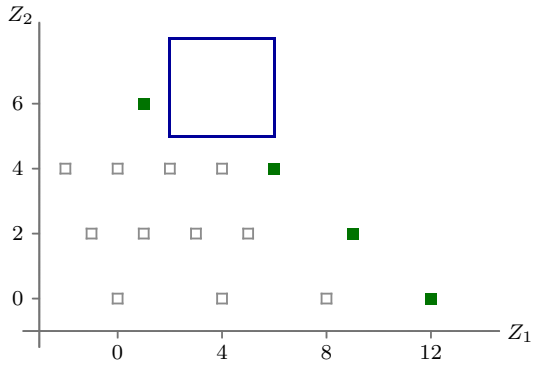
$Z_1 \geqslant 7$. $x_B = (2, 0, 0)$ with $\Phi = -1$, again dominated; the centre is $(3, 0, 0)$ at $Z = (12, 0)$ and Q gives $-5/4$, which improves the incumbent from $-7/4$. Two children follow.

Step 3 — a maximiser that is already efficient



$Z_2 \geqslant 5$, $Z_1 \leqslant 6$. Here $x_B = (0, 1, 2)$ with $\Phi = -1/5$ and the test returns $\theta = 0$: the maximiser *is* efficient. So it becomes the incumbent directly, and **no Q is owed** — x_B already maximises Φ over the whole box, hence over its own slice. This is the answer, though the search does not know that yet.

Step 4 — the early exit



The best bound still open is $-1/5$, which the incumbent already equals, so it cannot be beaten. Because the open list is ordered by bound descending, no other box can beat it either: the search stops and reports **four boxes still open and never solved**. Without this line all four would have been solved, one at a time, only to be dropped.

What the four steps showed. A repair (twice), a maximiser already efficient, a box dropped on its bound, and the early exit — every branch of Algorithm 1 in the companion *step by step* document, on fifteen feasible points.

4 What the two examples say together

Table 1: The two runs.

	n	$\lvert D\rvert$	$\lvert E(P_D)\rvert$	sub-problems	Φ^*
A (published instance)	2	11	7	5	5/17
B	3	15	4	4	$-1/5$

1. **The unconstrained maximiser of Φ is dominated in both.** $\Phi = 1$ at $(0, 0)$ in A against an answer of $5/17$; $\Phi = 9/5$ at $(0, 2, 0)$ in B against an answer of $-1/5$. The efficiency test is not a formality on these instances — it is the difference between an answer and a plausible wrong number.

2. **The optimum is reached through Q, not through a maximiser, in A.** $(3, 3)$ is found as the centre repaired to from the dominated $(1, 2)$, and its value comes from the Q sub-problem. An implementation that skipped Q — reasonable-looking, if one forgets that Φ is a function of x and not of $Z(x)$ — would miss it.

3. **Neither run enumerated the efficient set.** Five sub-problems for seven efficient points, four for four. On the instances of the main note the gap is the whole point: eight boxes settle an instance with 71 efficient points among 629 feasible ones.